Thrust Areas of Research (TAR)

TAR-1	Magnetic Fluids based technologies **(Dr. Kinnari Parekh)**
TAR-2	Development of inorganic and perovskite based solar cells **(Dr. Anjana Kothari)**
TAR-3	Bioremediation for water purifications, nuclear biology (**Dr. Kamesh Baskaran)**
TAR-4	Metal oxide Based NH_3 sensor in high humid atmosphere & metal dichalcogenides for photo detection (**Dr. Vanaraj Solanki)**
TAR-5	Biomimetic Model Studies of Metalloenzymes Active Site (**Dr. Atanu Banerjee)**
TAR-6	Magnetorheological fluids based technologies **(Prof. R V Upadhyay & Dipal Patel)**
TAR-7	Engineering Solutions to Biomedical Problem **(Prof. Chakradhar Iyyunni)**

TAR-1 Magnetic Fluids based technologies **(Dr. Kinnari Parekh)**

A. Magnetic Fluid Hyperthermia (MFH): (Dr Kinnari Parekh)

Target: Maximum SAR with 42.4 - 45.5°C temperature at minimum fluid dose (< 10 mg/ml) and within the safety limit of magnetic field.

1st year (2019-20)	**2nd year (2020-21)**	**3rd year (2021-22)**
a) The biocompatible TSMF will be tested for its cytotoxic effect in normal and cancerous cell lines. b) The cell death of the particles will be then tested with cancerous cell line using different protocols. The parameters like dose, frequency, amplitude, time will be optimized for subsequent experiments. **Deliverables:** **The results will be analyzed and will be sent for the publication in peer reviewed journal.**	a) Synthesis protocol for preparing samples with combined chemotherapeutic drug and TSMF fluid will be initiated. b) The samples will be characterized using physical, magnetic and hyperthermia techniques. **Deliverables:** **Final protocol will be tested for further research project.**	a) MNP conjugated with chemo drug will be tested for its cytotoxic effect on cancerous cell lines. b) Protocols for IC50 and other operating parameters will be optimized. Deliverables Based on the results obtained the in vivo experiments on animal can be planned.

B. Rotary shaft magnetic fluid seal: (Dr. Kinnari Parekh)

Target: To develop magnetic fluid seal with 100000 <DN < 200,000(D=shaft diameter and N=speed in rpm).

1st year	**2nd year**	**3rd year**
a) The custom designed magnetic fluid will be synthesized and characterized to test designed seal performance. b) The seal will be tested for its sealing capacity using the above synthesized fluid under variable rpm and number of stages. **Deliverables:** The results will be analyzed jointly and will be sent for the publication in peer reviewed journal.	a) High torque low vacuum rotary seal will be designed using FEMM. b) The appropriate magnetic fluid will be synthesized characterized and for the testing in prototype seal. c) If possible, the seal will be tested for its sealing capacity using the above synthesized fluid under variable rpm and number of stages. **Deliverables:** Industry collaboration will be established based on the results and design	a) Design and optimization of seal with 100000< DN< 200000 will be carried out. b) Based on the design parameters, the seal will be fabricated c) Testing & evaluation of the designed seal Deliverables MF seal with high torque.

C. Heat transfer device applications (Dr Kinnari Parekh)

1st year	**2nd year**	**3rd year**
a) A prototype set-up for heat transfer capacity will be designed & fabricated. b) Fluid will be tested for the heat transport. **Deliverables:** The results will be analyzed and if any modification required will be incorporated for further study. The publication in peer reviewed journal.	**Deliverables:** a) Based on the output of the experiments carried out in first two years, the novelty of our work will be proposed to submit at proposal for the funding. b) The results will be analyzed will be sent for the publication in peer reviewed journal.	a) CFD/COMSOL of coolant fluid parameters b) Optimization of parameters Deliverables Optimized coolant property.

D. Mono-disperse magnetic fluid (Dr Kinanri Parekh)

International collaboration: Professor Antonio Martins Neto, Professor of Physics at the Experimental Physics Department, Physics Institute, University of São Paulo - SP (Brazil)

1st year	**2nd year**	**3rd year**
a) The future experiments will be planned based on the results obtained from the preliminary characterizations done at India as well as at Brazil. **Deliverables:** • The data will be analyzed jointly and will be sent for the publication in peer reviewed journal. Based on the result new research proposal can be submitted for the funding.	**Deliverables:** • Based on the output of the experiments carried out in first two years, the work will be sent for the publication in peer reviewed journal.	• To develop protocol for dispersion of monodispersed particles in polar and non-polar carriers. **Deliverables** Monodispersed fluid in non-polar and polar medium.

TAR -2: Development of inorganic and perovskite based solar cells **(Dr. Anjana Kothari)**

1st year – Solar PV (2019-20)	**2nd year – Solar thermal**	**3rd year – Solar PV &Solar Thermal**
i. Inkjet printing (IJP) of inks on glass and conducting glass plates/substrates ii. Structural (XRD, SEM, AFM, TEM), Optical (UV-VIS: A/T/R) and electrical (photo-/conductivity, TEP, etc.) characterizations of inks and inkjet printer films iii. IJP of stack of thin films and its characterization iv. Measurement of solar cell parameters **Deliverables:** Two research papers will be published in the SCI journal	i. Design of prototype dryer for drying of vegetables. ii. Purchase of accessories/ consumables. iii. Fabrication of the dryer. iv. Preliminary data will be collected. v. These data will be used to form a research project proposal. **Deliverables:** A research project proposal will be submitted	i. Fabrication of solar PV module for CHARSUAT ii. Initiation of Development of solar thermal devices. **Deliverables** Solar PV-module Basic development of solar thermal device.

TAR -3: Bioremediation for water purifications, nuclear biology (**Dr. Kamesh Baskaran)**

Development of Lab on chip for heavy metal detection for food and beverages

1st year (2019-2020)	2nd year (2020-2021)	3rd year (2021-2022)
• Development of Lab on chip using microfluidics patterning on paper and other polymer substrate using 3D printer • Printing with FDM and then converted for DLP depending on availability with vendor	• Optimization of analyte-ligand-chromophore for various heavy metals • Interference study of analyte-ligand-chromophore for chip • Comparision of chip with ICP- • OES ad ICP-MS • Partial field analysis for evaluation of chip	• Field sample analysis in lab for environmental parameters • On-spot field analysis using chip and calculating daily intake and target hazard quotient for risk assessment

TAR -4: Metal oxide Based NH_3 sensor in high humid atmosphere & metal dichalcogenides for photo detection (**Dr. Vanaraj Solanki)**

1st year (2019-2020)	2nd year (2020-2021)	3rd year (2021-2022)
• Development of gas sensing facility • Photo-detection facility restoration • Synthesis of porous metal oxides and metal dichalcogenides	• Fine tuning of synthesis parameters to achieve desirable synthesis route • Grown materials will be characterized in terms of morphology, crystal structure, electric and electronic properties etc. • Device fabrication for gas sensing application • Device fabrication for photo-detection application	• Gas sensing and photo-detection responses will be tested for fabricated devices • Interpretation and correlation of different experimental results • Extension of the gas sensing towards artificial e-nose will be initiated

TAR -5: Biomimetic Model Studies of Metalloenzymes Active Site (**Dr. Atanu Banerjee)**

1st year (2019-2020)	2nd year (2020-2021)	3rd year (2021-2022)
• Set-up a synthetic laboratory. • Design and synthesis of the new sets of ligand system which will mimic the active site of the metalloenzymes having 3 and 4 Histidine core. • Generate the metalloenzyme active site by the complexsation of the ligands with the metal centre.	• Fine tuning of the coordination geometry of the metalloenzyme active site by introducing/varying the steric or electronic bulk into ligand system. • Crystallographic, spectroscopic and electrochemical characterization of the synthetic models. • Substrate binding with the model active site and verifying the efficacy of the system in bioremediations.	• Dioxygenase study, isolation and characterization of the intermediates (if appear). • Isotope labeling (O^{18}, $H_2O_2^{18}$) experiment. • DFT Calculation and the interpretation of the experimental Data.

TAR -6: Magnetorheological fluids based technologies **(Prof. R V Upadhyay& Dipal Patel)**

A. MR-fluid Damper design and fabrication

1st year – Fluid synthesis	**2nd year – Damper device design**	**3rd year – commercialization**
1. Damper design will be defined. 2. Optimization of damper design. 3. Fabrication of damper as per design, 4. Validation of damper design using MR fluid. 5. Finalizing the fluid and damper properties.	1. Prototype model will be fabricated. 2. Industry will be contacted for use of the damper. 3. Manufacturing at large scale in collaboration with industry	Identifying the industries and end user for commercialization of the product.

B. MR-fluid damper for Knee joint: design and fabrication (Dr R V Uapdhyay, Dr Bala Ganpati & Dipal Patel)

1st year – Knee joint understanding	**2nd year – Knee device design**	**3rd year – commercialization**
1. Physiotherapy department help will be sought to understand biomechanics. 2. A design will be formulated for the required Knee design. 3. MR fluid will be synthesized for this application. 3. The fluid will be characterized and optimized.	1. Design will be defined. 2. Optimization of design. 3. Fabrication as per design, 4. Validation of r design using MR fluid. 5. Finalizing the fluid and knee design.	1. Prototype model will be fabricated. 2. Industry will be contacted for use of the same 3. Manufacturing at large scale will be initiated with the support of DST.

C. Projects applied (with title): ____03_____

Sr. No.	Name of Project	Amount	Funding Agency	Duration	Applied date	Status
1.	Synthesis and Study on biologically relevant model compounds for the active site of Metalloenzy-mes **PI:** Dr. Atanu Banerjee (CHARUSAT) **CO-PI:** Prof. Prafulla Jha (M S University of Baroda)	Rs. 28,86,000	SERB- CRG-DST	3 years	7th June, 2019	Project under evaluation
2.	Development of Lab on chip (LOC) for rapid detection of heavy metal contaminants in Food and beverages. **PI**-Dr Kamesh Viswanathan Baskaran	Rs. 30,93,120	SERB-CRG-DST	3 years	8th June, 2019	Project under evaluation
3	Studies on Metal Dichalcogenide and Porous Metal Oxide based Hybrid Nanostructures for Multi-functional Applications **PI**-Dr. Vanarajsinh Solanki	Rs. 33,00,000	SERB-DST-SRG	2 years	19th April, 2019	Project under evaluation

D. Employee wise details of Publications (list) (year 2018-19)

Please see Annexure A

E. Research Highlights and Special academic Achievements

1. Mr. Dhaval Vankhade had awarded Inspire Fellowship from Department of Science and Technology, Government of India. (2015 - 2018).
2. Dr. Kinnari Parekh visited University of Sao Paulo, Sao Paulo, Brazil as a visiting scientist for 1st Aug 2018 to 30th Sep 2018.

F. Consultancy / other income earned (if any). Give details:

1. Consultancy through selling of magnetic fluid to other users as per their demand. Rs. **20,000**/-
2. Equipment usage charges earned through our various sophisticated equipment facilities for outside users. Rs. **2,15,275**/- (July 2018-June 2019).

G. Employee wise Patent:

1. Dr. Kinnari Parekh published a complete Indian Patent (No. 2164/MUM/2014).
2. Dr. Kinnari Parekh published a PCT Patent (WO 2016/001939 A2)

H. Financial details:

Year	Income	Expenditure
2018-19	17420313	9890796

(Account section shall help.)